ANAYA JAIN

Aspiring AI and Computer Science Researcher

https://anayajain.me/ Pune, India

SUMMARY

Grade 12 student with a strong foundation in Computer Science, AI, and Mathematics. Committed to expanding access to STEM education through teaching, mentorship, and community-building. Seeking to pursue Computer Science and AI at the undergraduate level, with an emphasis on building technology and doing research that creates meaningful impact.

EDUCATION

Grade 12 (CBSE) *Apr '27*

EON Gyanankur English School, Pune, India

Grade 9, 10 (CISCE) *Apr '25*

Vibgyor High, Magarpatta, Pune, India

Final Grade: 97%

SKILLS

Programming Languages & Technologies: Python, C, Java, JavaScript, SQL, HTML, CSS, TensorFlow, Deep Learning, CNNs, DenseNet-121, Image Classification, Grad-CAM, Data Analysis, Flask, Responsive Web Design, ThingSpeak, ESP32, GitHub, Sensor Integration, Data Structures, Algorithms, Object-Oriented Programming, Database Management, Mathematical Problem-Solving

Coursework & Certifications: Harvard CS50x: Introduction to Computer Science; AP Computer Science A; AP Calculus AB

Analytical Skills: Algorithms, Data Structures, Mathematical Problem Solving, Data Analysis

ACADEMIC and RESEARCH PROJECTS

Rice Leaf Disease Detection using Deep Learning *June '25 - Present*

Research Poster Presenter, IEEE MIT URTC 2025

- Analyzed lesion severity across 10,407 rice leaf images using a DenseNet-121 model paired with Grad-CAM heatmaps to identify and quantify diseased regions.
- Presented the work as a poster at the IEEE MIT Undergraduate Research Technology Conference (URTC) 2025.

Research Author, Journal of Emerging Investigators

- Proposed integrating the approach into drone-based precision agriculture systems to support scalable, early-intervention disease monitoring for farms.
- Paper accepted in Journal of Emerging Investigators (JEI), passed scientific review, and is under copyediting

Air Atlas: Hyperlocal Air Quality Mapping *Mar '26 – May '26*

IoT-based environmental sensing project developed under CERN mentorship

https://air-atlas.netlify.app/

- Built and deployed an ESP32-based IoT air-quality monitoring node in Pune, measuring PM1, PM2.5, PM10, temperature, humidity, and atmospheric pressure.
- Monitored sensor readings through ThingSpeak for one month, collecting continuous data from the Pune node.
- Contributed Pune node data to Air Atlas, a live dashboard that aggregates measurements from 10 locations across India and visualizes AQI trends, pollution spikes, temperature, humidity, and other environmental patterns.
- Collaborated with team members operating nodes in different locations to create a hyperlocal, multi-location view of air quality across India.

Guitar Chord Shape Identifier *Mar '26*

https://rlr2vx.csb.app/

Kode with Klossy CODE-A-BRATION project

- Built a website and developed an image classification model to identify guitar chord shapes, helping beginner guitarists visualize correct finger placement.

Learning Buddy *June '25 - Oct '25*

Harvard's CS50x final project

- Developed a web platform connecting underserved students with volunteer tutors using HTML, CSS, JavaScript, Python (Flask), and SQLite.
- Applied CS50x concepts, including data structures, algorithms, database management, and backend development, to implement user registration, tutor matching, and session scheduling.

EXPERIENCE

New York Academy of Sciences – Junior Academy Member *Sep '26 – Present*

- Selected for the Junior Academy, a global community connecting high school students with STEM opportunities, mentorship, and research experiences.
- Collaborating on a six-person team for NeuroPulse, a wearable neurotechnology project exploring how ear-EEG, PPG, and HRV signals integrate to reveal patterns associated with atrial fibrillation (AFib).
- Exploring the software side of the project, focusing on analyzing multimodal physiological signals and their potential applications in long-term, non-invasive AFib monitoring.

Kode With Klossy Summer Camp - Web Development *Aug '26*

- Completed an intensive web development program focused on HTML, CSS, JavaScript, and responsive web design.
- Collaborated with a team to create a website exploring the intersection of art and education, highlighting statistics on women's representation and advocating for women's rights through digital storytelling.

Girls Who Code Summer Pathways - AI & Data Science; AI+Society *Jun '26 – Aug '26*

- Engaged in comprehensive studies of data analytics and intelligent systems through practical software projects
- Investigated the intersection of technology and ethics, focusing on the global effects of artificial intelligence.

Leaf Independent Learner *Apr '26 - May '26*

- Selected as an Independent Learner in Leaf's five-week program, independently studying The Mathematics of Morality and Dilemmas and Dangers in AI through self-directed readings and coursework.
- Explored moral decision-making, mathematical approaches to ethics, AI alignment and safety, AI risks, and the societal implications of increasingly capable AI systems, developing independent perspectives on complex ethical questions.

HERizon: Founder and Lead *Mar '26 – Present*

https://herizonadmin.wixsite.com/herizon

- Founded a student-led initiative to empower girls in STEM, reaching 100+ students to date.
- Organized monthly webinars with industry leaders and collaborated with organizations across India and Kenya to expand outreach.

Schoolhouse.world: Volunteer Tutor Math & President, Dialogues Club *Jan '26 – Present*

Free Online Tutoring — Schoolhouse

- Hosted 22 SAT Math, Pre-Algebra, and Calculus sessions, impacting 113 learners across 25 countries and logging 1,200+ tutoring minutes.
- Conducted live SAT Math Bootcamps and tutoring sessions in Pre-Algebra and Calculus.

- Served as Summer and Fall Dialogues Club President.

Harvard CS50x: Introduction to Computer Science *June '25 – Oct '25*

- Introduction to Computer Science, building a foundation in algorithms, data structures, and problem-solving through nine programming problem sets in C, Python, and SQL.
- Built a web platform connecting underprivileged students with volunteer tutors, using HTML, CSS, JavaScript, and Python (Flask) with a SQL database to handle user sign-ups, tutor matching, and session scheduling.

Lodha Genius Programme Scholar, Ashoka University *May '25 – Dec '25*

- Selected among 8,000 applicants nationwide, placing in the top 4% of students selected in the country.
- Completed advanced coursework in Pure Mathematics, including Number Theory, Linear Algebra, Geometry, and Group Theory.
- Completed the online continuing-learning module in the five-month Physics of Stars course; selected as one of the top 5 students in my course to visit NCRA-TIFR and the Giant Metrewave Radio Telescope (GMRT).
- Selected as a participant in the CERN International Masterclass, analyzing real particle physics data and studying experiments such as ALICE, CMS, and Antimatter Factory; attended Nobel laureate seminars; led 20 students as Ambassador.

Vibgyor High: Student Council Editor *Jun '23 – Mar '24*

- Led content planning, editing, and publishing for the school's official magazine publication.
- Developed strong communication and leadership skills through team coordination and editorial decision-making.

Model United Nations – Intermediate Level *Jun '23 – Mar '24*

- Represented China in the UN General Assembly at VIBGYOR MUN, debating the Russia-Ukraine war and researching China's position on the conflict; earned a Special Mention for diplomacy, research, and public speaking.

Vibgyor High: Social Service Responsibility Cell *Jun'23–Mar '24*

- Organized and led community outreach initiatives focused on education, raising over ₹7000 to support underprivileged students.
- Participated in donation drives and awareness campaigns, mentored junior members, and promoted social responsibility among peers.

INTERESTS & EXTRACURRICULAR ACTIVITIES

John Locke Institute 2026 Global Essay Prize - Commendation *April – Oct 2026*

- Awarded Commendation in Science & Technology among 100,227 entries for "Is free speech the enemy of science?"; invited to the 2026 Awards Dinner in London.

Trinity College London – Acoustic Guitar *Mar'21 – Present*

- Completed Initial, Grade 1, and Grade 3 with distinction; currently preparing for Grade 5 and building strong musical skills and discipline through structured training and examinations.

LifeTide Blog: https://lifetide01.wordpress.com/ *2025 - Present*

- Publish reflective essays exploring science, curiosity, and personal growth.

School Dance Team *2023 – 2024*

- Led the senior secondary dance team.
- Participated in and choreographed school performances and annual productions.